

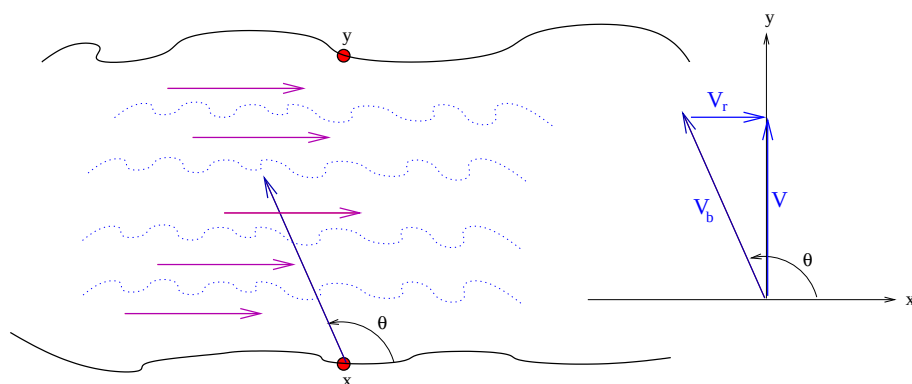
DEPARTMENT OF MATHEMATICS, IIT - GUWAHATI

Odd Semester of the Academic year 2022 - 2023

MA 101 Mathematics I

Problem Sheet 1: Revision of vectors, equations of lines and planes, vector differentiation, limits and continuities of functions of several variables.

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1.

A man wants to paddle his boat across a river from point X to the point Y on the opposite shore directly across from X . If he can paddle the boat at the rate of 5 kilometers per hour and the current in the river is 3 kilometers per hour, in what direction θ should he steer his boat in order to go straight across the river? Also what is his resultant speed across the river?

Solution: Let $\vec{V}_b$ be the velocity of the boat and $\vec{V}_r$ be the velocity of the river and let $\vec{V}$ be the resultant velocity of the boat across the river.

Clearly if $\vec{V}_r = 3\mathbf{i}$ then $\vec{V}_b = 5\cos(\theta)\mathbf{i} + 5\sin(\theta)\mathbf{j}$ and $\vec{V} = \alpha\mathbf{j}$ for some $\alpha > 0$, where $\alpha = |\vec{V}|$ is the relative speed of the boat across the river.

$$\text{Also } \vec{V}_b + \vec{V}_r = \vec{V} \Rightarrow (5\cos(\theta) + 3)\mathbf{i} + 5\sin(\theta)\mathbf{j} = \alpha\mathbf{j} \quad (1)$$

Equating the components of the vectors on both sides of equation (1) we get $5\cos(\theta) + 3 = 0 \Rightarrow \cos(\theta) = -\frac{3}{5} \Rightarrow \theta = 2.214$ radians, or $\theta = 126.87^\circ$.

Since $\alpha^2 + 3^2 = 5^2$, $\alpha = 4$, hence the resultant speed is 4 kilometers per hour.

2. If $\mathbf{a} = \langle 3, 0, -2 \rangle$, find a vector $\mathbf{b}$ such that $\text{comp}_{\mathbf{a}}\mathbf{b} = 2$.

Solution: Let $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, where $b_1, b_2, b_3 \in \mathbf{R}$. We know that

$$\text{comp}_{\mathbf{a}}\mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|}.$$

Since $\mathbf{a} = \langle 3, 0, -2 \rangle$ and $\text{comp}_{\mathbf{a}}\mathbf{b} = 2$, we get that

$$\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|} = 2.$$

This implies that

$$3b_1 - 2b_3 = 2\sqrt{13}$$

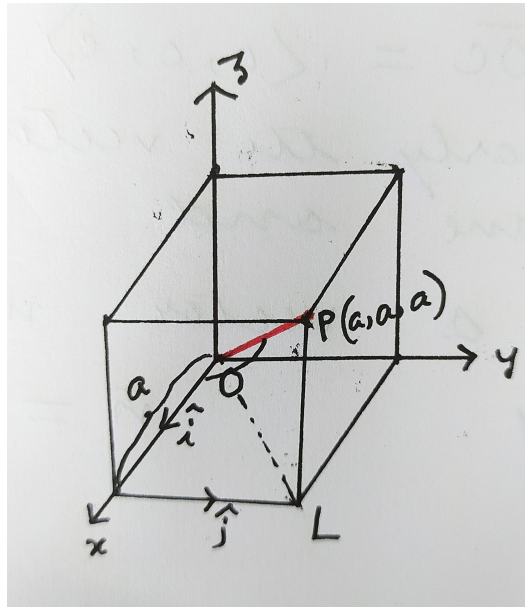
$$\Rightarrow b_3 = \frac{3}{2}b_1 - \sqrt{13}.$$

We may choose $\mathbf{b} = \langle 0, 0, -\sqrt{13} \rangle$ or $\mathbf{b} = \langle b_1, b_2, \frac{3}{2}b_1 - \sqrt{13} \rangle$ such that $b_1, b_2 \in \mathbf{R}$.

3. Find the angle between the diagonal of a cube and one of its edges. What about the angle between the diagonal of a cube and the diagonal of one of its faces?

Solution: For simplicity, we may assume the cube to lie in the first octant and each of its edges of length a .

Join the origin, which is one of the intersection point of the three edges to its op-



posite point P of intersection of another three edges such that $\overrightarrow{OP}$ forms a diagonal. Clearly then $\overrightarrow{OP} = \langle a, a, a \rangle$ so that if α is the angle between the edge along x -axis and the diagonal, then

$$\overrightarrow{OP} \cdot \hat{i} = \langle a, a, a \rangle \cdot \langle 1, 0, 0 \rangle$$

$$\Rightarrow |\overrightarrow{OP}| |\hat{i}| \cos \alpha = a$$

$$\Rightarrow \sqrt{3}a \cos \alpha = a \Rightarrow \alpha = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right).$$

For the second part, one can see that the diagonal of one of the faces can be taken as the vector $\overrightarrow{OL} = \langle a, a, 0 \rangle$ so that if β is the angle between $\overrightarrow{OL}$ and $\overrightarrow{OP}$, then as before, we have:

$$\overrightarrow{OP} \cdot \overrightarrow{OL} = \langle a, a, a \rangle \cdot \langle a, a, 0 \rangle$$

$$\Rightarrow \sqrt{3}a \sqrt{2}a \cos \beta = 2a^2$$

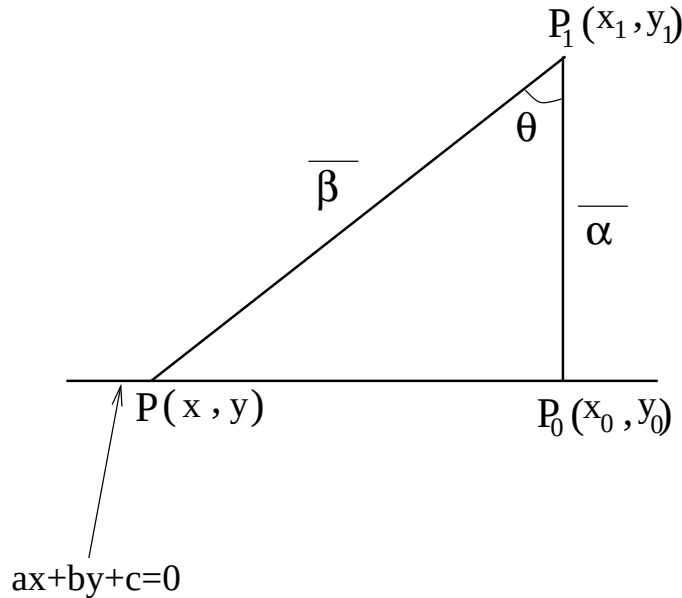
$$\Rightarrow \beta = \cos^{-1} \left(\sqrt{\frac{2}{3}} \right).$$

4. Use a scalar projection to show that the distance from a point $P_1(x_1, y_1)$ to the line $ax + by + c = 0$ is

$$\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}.$$

Use this formula to find the distance from the point $(-2, 3)$ to the line $3x - 4y + 5 = 0$.

Solution:



Let $P_0(x_0, y_0)$ be the foot of the perpendicular vector from $P_1(x_1, y_1)$ to the line $ax + by + c = 0$,

then distance of $P_1(x_1, y_1)$ from the line $ax + by + c = 0$ is given by $|\overrightarrow{P_1P_0}|$.

Let $P(x, y)$ be a point on the line $ax + by + c = 0$,

then $P_0(x_0, y_0)$ and $P(x, y)$ satisfy the equations $ax_0 + by_0 + c = 0$ and $ax + by + c = 0$, respectively

$$\Rightarrow a(x - x_0) + b(y - y_0) = 0,$$

$$\Rightarrow \langle a, b \rangle \cdot \langle x_0 - x, y_0 - y \rangle = 0.$$

Hence $\langle a, b \rangle \perp \langle x_0 - x, y_0 - y \rangle$ (or $\langle a, b \rangle$ is orthogonal to $\langle x_0 - x, y_0 - y \rangle$)

$$\Rightarrow \langle a, b \rangle \perp \overrightarrow{P_1P_0}.$$

But $\overrightarrow{P_1P_0} \perp \overrightarrow{PP_0}$ therefore any nonzero vector $\bar{\alpha}$ along $\overrightarrow{P_1P_0}$ must be parallel to $\langle a, b \rangle$, and $\bar{\alpha} = \lambda \langle a, b \rangle$ for some $\lambda \in \mathbf{R}$, $\lambda \neq 0$.

If we denote $\overrightarrow{P_1P}$ by $\bar{\beta}$, then

$$|\overrightarrow{P_1P_0}| = |\text{proj}_{\bar{\alpha}}(\bar{\beta})| = \frac{|\bar{\alpha} \cdot \bar{\beta}|}{|\bar{\alpha}|}$$

$$= \left| \frac{\lambda(a(x_1 - x) + b(y_1 - y))}{|\lambda| \sqrt{a^2 + b^2}} \right| = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}.$$

Using the above formula the distance from $(-2, 3)$ to the line $3x - 4y + 5 = 0$ is given by $\frac{|3(-2) + (-4)3 + 5|}{\sqrt{3^2 + (-4)^2}} = \frac{13}{5}$.

5. (a) Find a point at which the given lines intersect:

$$\mathbf{r}_1(t) = \langle 1, 1, 0 \rangle + t \langle -1, 1, 2 \rangle, \quad t \in \mathbf{R}$$

$$\mathbf{r}_2(s) = \langle 2, 0, 2 \rangle + s \langle -1, 1, 0 \rangle, \quad s \in \mathbf{R}$$

- (b) Find the equation of the plane that contains these lines.

Solution: (a) We can denote the lines as

$$\mathbf{r}_1(t) = \langle 1 - t, 1 + t, 2t \rangle, \quad t \in \mathbf{R}.$$

$$\mathbf{r}_2(s) = \langle 2 - s, s, 2 \rangle, \quad s \in \mathbf{R}.$$

If the lines intersect then there exists $t, s \in \mathbf{R}$ such that $2 - s = 1 - t$, $s = 1 + t$ and $2t = 2$, which implies $t = 1$ and $s = 2$ and the lines intersect at $P_0 = (0, 2, 2)$.

(b) The direction vectors of the two lines are given by $\langle -1, 1, 2 \rangle$ and $\langle -1, 1, 0 \rangle$.

A normal to the plane containing these two lines is given by

$$\langle -1, 1, 2 \rangle \times \langle -1, 1, 0 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 1 & 2 \\ -1 & 1 & 0 \end{vmatrix} = -2\mathbf{i} - 2\mathbf{j} = \langle -2, -2, 0 \rangle.$$

Hence equation of the required plane (which passes through the point $(0, 2, 2)$) is given by

$$-2(x - 0) - 2(y - 2) + 0(z - 2) = 0$$

$$\text{or } x + y = 2.$$

6. Find a formula for the distance D from a point $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$.

Solution:

Let $P_0(x_0, y_0, z_0)$ be the foot of the perpendicular vector from $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$, then $\overrightarrow{P_1P_0}$ is a normal vector $\bar{n}$ to the plane and the distance of $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$ is given by $|\overrightarrow{P_1P_0}|$.

Let $P(x, y, z)$ be a point on the plane $ax + by + cz + d = 0$, since $P_0(x_0, y_0, z_0)$ also lies on the plane $ax + by + cz + d = 0$,

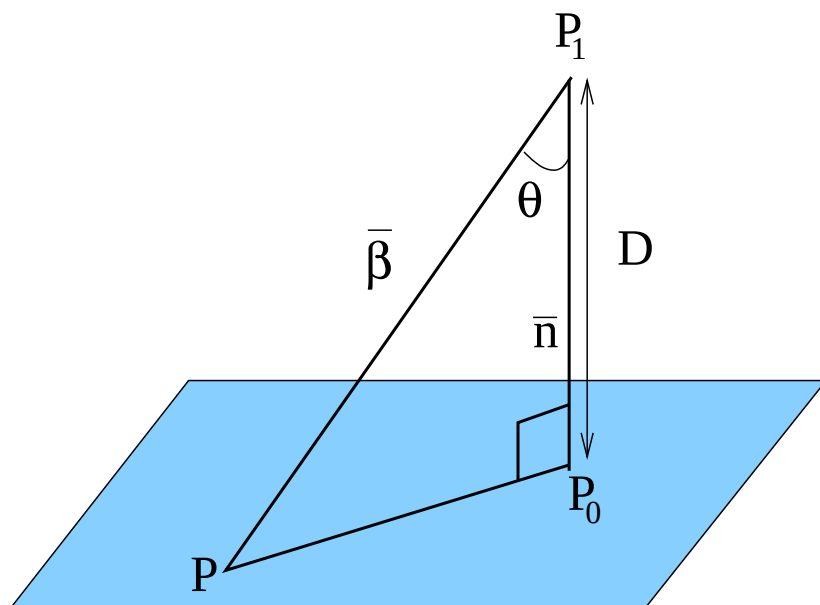
$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

Therefore any nonzero normal vector $\bar{n}$ of the plane given by $\bar{a}$ is of the form

$$\bar{a} = \lambda \langle a, b, c \rangle \text{ for some } \lambda \in \mathbf{R}, \lambda \neq 0.$$

If we denote $\overrightarrow{P_1P}$ by $\bar{\beta}$, then

$$|\overrightarrow{P_1P_0}| = |\text{proj}_{\bar{a}}(\bar{\beta})| = \frac{|\bar{a} \cdot \bar{\beta}|}{|\bar{a}|}$$

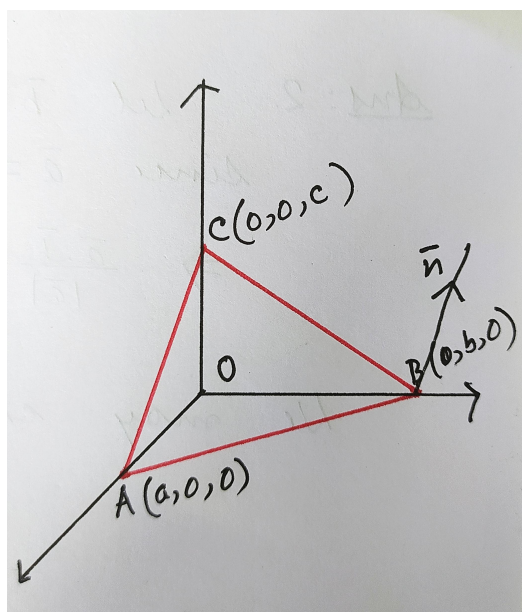


$$= \left| \frac{\lambda(a(x_1 - x) + b(y_1 - y) + c(z_1 - z))}{|\lambda|\sqrt{a^2 + b^2 + c^2}} \right| = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

7. Find the equation of a plane with x -intercept a , y -intercept b and z -intercept c such that $a, b, c \neq 0$.

Solution: A part of the plane is shown in the figure below.

If $A(a, 0, 0)$, $B(0, b, 0)$ and $C(0, 0, c)$ are the points where the plane intercepts the



x -, y - and z - axes respectively, then

$$\overrightarrow{OA} = \langle a, 0, 0 \rangle$$

$$\begin{aligned}\overrightarrow{OB} &= \langle 0, b, 0 \rangle \\ \overrightarrow{OC} &= \langle 0, 0, c \rangle.\end{aligned}$$

Clearly the vectors $\overrightarrow{AB}$ and $\overrightarrow{BC}$ lie on the plane and $\overrightarrow{AB} = \langle -a, b, 0 \rangle$ and $\overrightarrow{BC} = \langle 0, -b, c \rangle$. Therefore, a vector normal to the plane is given by

$$\vec{n} = \overrightarrow{AB} \times \overrightarrow{BC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -a & b & 0 \\ 0 & -b & c \end{vmatrix} = \langle bc, ca, ab \rangle.$$

Since $A(a, 0, 0)$ is a point on the plane, therefore the required equation is

$$bc(x - a) + ca(y - 0) + ab(z - 0) = 0,$$

that is,

$$bcx + cay + abz = abc,$$

or

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1.$$

8. Find a point on the curve $\mathbf{r}(t) = 4\cos(t)\mathbf{i} + 4\sin(t)\mathbf{j} + 3t\mathbf{k}$ at a distance 10π units from the point $(4, 0, 0)$ along the curve in the direction of increasing arc length.

Solution: The point $(4, 0, 0)$ corresponds to $t = 0$.

Since $\mathbf{r}(t) = 4\cos(t)\mathbf{i} + 4\sin(t)\mathbf{j} + 3t\mathbf{k}$ is of the form

$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where $f(t), g(t), h(t)$ are continuously differentiable functions in t ,

$$\mathbf{r}'(t) = -4\sin(t)\mathbf{i} + 4\cos(t)\mathbf{j} + 3\mathbf{k}$$

and the arc length $L(t_1)$ of $\mathbf{r}(t)$ from $t = 0$ to $t = t_1$ is given by:

$$L(t_1) = \int_0^{t_1} |\mathbf{r}'(t)| dt = \int_0^{t_1} \sqrt{16(\sin^2(t) + \cos^2(t)) + 9} dt = 5t_1.$$

If $L(t_1) = 5t_1 = 10\pi$, then $t_1 = 2\pi$, and the corresponding point on the curve is given by:

$$\mathbf{r}(2\pi) = 4\cos(2\pi)\mathbf{i} + 4\sin(2\pi)\mathbf{j} + 3(2\pi)\mathbf{k} = 4\mathbf{i} + 6\pi\mathbf{k}.$$

9. Reparametrize the curve

$$\mathbf{r}(t) = \left(\frac{2}{t^2 + 1} - 1 \right) \mathbf{i} + \frac{2t}{t^2 + 1} \mathbf{j}$$

with respect to the arc length measured from the point $(1, 0)$ in the direction of increasing t . Express the parametrization in its simplest form. What can you conclude about the curve?

Solution: Since

$$\mathbf{r}(t) = \left(\frac{2}{t^2 + 1} - 1 \right) \mathbf{i} + \frac{2t}{t^2 + 1} \mathbf{j},$$

the point $(1, 0)$ corresponds to $t = 0$.

Also $f(t) = \frac{2}{t^2 + 1} - 1$ and $h(t) = \frac{2t}{t^2 + 1}$ are continuously differentiable at all $t \in \mathbf{R}$.

$$\mathbf{r}'(t) = \left(\frac{-4t}{(t^2 + 1)^2} \right) \mathbf{i} + \left(\frac{-2t^2 + 2}{(t^2 + 1)^2} \right) \mathbf{j}, \text{ and}$$

$$|\mathbf{r}'(t)| = \sqrt{\left(\frac{-4t}{(t^2+1)^2}\right)^2 + \left(\frac{-2t^2+2}{(t^2+1)^2}\right)^2} = \frac{2}{t^2+1}.$$

$$\text{Hence } s(t) = \int_0^t |\mathbf{r}'(\xi)| d\xi = \int_0^t \frac{2}{\xi^2+1} d\xi = 2 \tan^{-1}(t)$$

$$\Rightarrow t(s) = \tan\left(\frac{s}{2}\right), \text{ for } 0 \leq s < \pi.$$

$$\mathbf{r}(t(s)) = \left(\frac{2}{(\tan(\frac{s}{2}))^2+1} - 1\right) \mathbf{i} + \frac{2 \tan(\frac{s}{2})}{(\tan(\frac{s}{2}))^2+1} \mathbf{j} = \cos(s) \mathbf{i} + \sin(s) \mathbf{j} \quad 0 \leq s < \pi.$$

Hence with this reparametrization, the points on the curve represent points on the upper half part of the unit circle centered at the origin, excluding the point $(-1, 0)$.

10. At what point does the curve $y = e^x$ have maximum curvature? What happens to the curvature as $x \rightarrow \infty$?

$$\textbf{Solution: } \kappa(x) = \frac{|y''(x)|}{(1 + (y'(x))^2)^{\frac{3}{2}}} = \frac{e^x}{(1 + e^{2x})^{\frac{3}{2}}}.$$

$$\Rightarrow \kappa'(x) = \frac{e^x(1 - 2e^{2x})}{(1 + e^{2x})^{\frac{5}{2}}}.$$

$$\Rightarrow \kappa'(x) = 0 \text{ if and only if } 1 - 2e^{2x} = 0 \text{ or } x = -\frac{1}{2} \ln(2).$$

Note that for $x < -\frac{1}{2} \ln(2)$, $\kappa'(x) > 0$ which implies

$\kappa(x)$ is a strictly increasing function in $(-\infty, -\frac{1}{2} \ln(2))$.

For $x > -\frac{1}{2} \ln(2)$, $\kappa'(x) < 0$ which implies

$\kappa(x)$ is a strictly decreasing function in $(-\frac{1}{2} \ln(2), \infty)$.

Hence the point of the curve at which curvature is maximum is given by

$$\left(-\frac{1}{2} \ln(2), e^{-\frac{1}{2} \ln(2)}\right) = \left(-\frac{1}{2} \ln(2), \frac{1}{\sqrt{2}}\right).$$

Note that $e^x < 1 + e^{2x}$, for all $x \in \mathbb{R}$,

$$\Rightarrow 0 \leq \kappa(x) = \frac{e^x}{(1 + e^{2x})^{\frac{3}{2}}} \leq \frac{1}{\sqrt{(1 + e^{2x})}} \quad (1)$$

$$\text{Since } \lim_{x \rightarrow \infty} \frac{1}{\sqrt{(1 + e^{2x})}} = 0, \quad (1) \quad \Rightarrow \quad \lim_{x \rightarrow \infty} \kappa(x) = 0.$$

11. Find the unit tangent vector, unit normal vector and the binormal vector of the curves at the corresponding points given below:

$$(a) \quad \mathbf{r}(t) = \left\langle t^2, \frac{2}{3}t^3, t \right\rangle, \quad \left(1, \frac{2}{3}, 1\right)$$

$$(b) \quad \mathbf{r}(t) = \langle \cos(t), \sin(t), \ln(\cos(t)) \rangle, \quad (1, 0, 0).$$

$$\textbf{Solution: } (a) \quad \mathbf{r}(t) = \left\langle t^2, \frac{2}{3}t^3, t \right\rangle,$$

$\Rightarrow \mathbf{r}'(t) = \langle 2t, 2t^2, 1 \rangle$, hence the unit tangent vector at $\mathbf{r}(t)$ of the curve is given by:

$$\Rightarrow \mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \left\langle \frac{2t}{2t^2+1}, \frac{2t^2}{2t^2+1}, \frac{1}{2t^2+1} \right\rangle.$$

Since the point $\left(1, \frac{2}{3}, 1\right)$ of the curve $\mathbf{r}(t)$ corresponds to $t = 1$, the the unit tangent vector at that point is given by:

$$\mathbf{T}(1) = \left\langle \frac{2}{3}, \frac{2}{3}, \frac{1}{3} \right\rangle.$$

$$\text{Also } \mathbf{T}'(t) = \left\langle \frac{2-4t^2}{(2t^2+1)^2}, \frac{4t}{(2t^2+1)^2}, \frac{-4t}{(2t^2+1)^2} \right\rangle.$$

$$\text{Since } \mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}, \mathbf{N}(1) = \frac{\mathbf{T}'(1)}{|\mathbf{T}'(1)|} = \left\langle -\frac{1}{3}, \frac{2}{3}, -\frac{2}{3} \right\rangle, \text{ and}$$

$$\mathbf{B}(1) = \mathbf{T}(1) \times \mathbf{N}(1) = \left\langle -\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \right\rangle.$$

(b) Since the point $(1, 0, 0)$ of the curve $\mathbf{r}(t)$ corresponds to $t = 0$, by following the same procedure as in part(a) we get:

$$\mathbf{T}(0) = \langle 0, 1, 0 \rangle$$

$$\mathbf{N}(0) = \left\langle -\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \right\rangle$$

$$\mathbf{B}(0) = \left\langle -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right\rangle.$$

12. The helix $\mathbf{r}_1(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k}$ intersects the curve $\mathbf{r}_2(t) = (1+t)\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$ at the point $(1, 0, 0)$. Find the angle of intersection of these curves.

Solution: The angle of intersection of these curves at $(1, 0, 0)$ is the angle between the tangents of these two curves at the point $(1, 0, 0)$.

$$\mathbf{r}_1(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k} \Rightarrow \mathbf{r}'_1(t) = -\sin t \mathbf{i} + \cos t \mathbf{j} + \mathbf{k}.$$

$$\text{Also } \mathbf{r}_2(t) = (1+t)\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k} \Rightarrow \mathbf{r}'_2(t) = \mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}.$$

Since the point $(1, 0, 0)$ corresponds to $t = 0$ for both $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$, if θ is the required angle then

$$\cos(\theta) = \frac{\mathbf{r}'_1(0) \cdot \mathbf{r}'_2(0)}{|\mathbf{r}'_1(0)| |\mathbf{r}'_2(0)|} = 0, \text{ or } \theta = \frac{\pi}{2}.$$

13. (a) A particle moves with constant speed along a curve in space. Show that its velocity and acceleration vectors are always perpendicular.
 (b) Let $\mathbf{r}(t) = (2t^3 + 3)\mathbf{i} + (\ln t)\mathbf{j} + 3\mathbf{k}$ be the position vector of a moving particle at time $t > 0$. Find the time(s) at which velocity and acceleration vectors are perpendicular.

Solution: (a) If $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ denotes the above curve, then its velocity, speed, acceleration at any time $t > 0$ is given by

$$\mathbf{r}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k}, \quad |\mathbf{r}'(t)| = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} \text{ and } \\ \mathbf{r}''(t) = x''(t)\mathbf{i} + y''(t)\mathbf{j} + z''(t)\mathbf{k}, \text{ respectively.}$$

Since speed is constant for the moving particle, $(x'(t))^2 + (y'(t))^2 + (z'(t))^2 = c$ for all t in the domain of the curve,

$$\Rightarrow 2(x'(t)x''(t) + y'(t)y''(t) + z'(t)z''(t)) = 0.$$

$$\Rightarrow \langle x'(t), y'(t), z'(t) \rangle \cdot \langle x''(t), y''(t), z''(t) \rangle = \mathbf{r}'(t) \cdot \mathbf{r}''(t) = 0, \text{ or } \mathbf{r}'(t) \perp \mathbf{r}''(t).$$

$$(b) \mathbf{r}'(t) = (6t^2)\mathbf{i} + \left(\frac{1}{t}\right)\mathbf{j} \text{ and } \mathbf{r}''(t) = (12t)\mathbf{i} + \left(-\frac{1}{t^2}\right)\mathbf{j}, \text{ for } t > 0.$$

$$\mathbf{r}'(t) \cdot \mathbf{r}''(t) = 0 \Rightarrow 72t^3 - \frac{1}{t^3} = 0, \Rightarrow t^6 = \frac{1}{72} \text{ or } t = \sqrt[6]{\frac{1}{72}}.$$

14. Find the limit if it exists, or show that the limit does not exist.

$$(a) \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$$

$$(b) \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}}$$

$$(c) \lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x^2 + y^2}$$

$$(d) \lim_{(x,y) \rightarrow (0,0)} \frac{xy^3}{x^2 + y^6}$$

$$(e) \lim_{(x,y) \rightarrow (0,0)} \frac{x \sin(x^2 + y^2)}{x^2 + y^2}$$

$$(f) \lim_{(x,y) \rightarrow (4,\pi)} x^2 \sin\left(\frac{y}{x}\right)$$

$$(g) \lim_{(x,y) \rightarrow (0,1)} f(x, y),$$

$$\text{where } f(x, y) = \begin{cases} \frac{x+y-1}{\sqrt{x-1}\sqrt{1-y}} & \text{if } x+y \neq 1 \\ 0 & \text{if } x+y = 1. \end{cases}$$

Solution: (a) $f(x, y) = \frac{xy}{x^2 + y^2} = \frac{m}{1 + m^2}$, for all (x, y) such that $y = mx$, $x \neq 0$,

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} = \frac{m}{1 + m^2} \text{ along } y = mx,$$

which is different for different straight lines passing through the origin.

hence $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ does not exist.

(For a more detailed discussion refer to (**) below).

((**) For all (x, y) on the line $y = 0$, $f(x, y) = f(x, 0) = 0$.

Similarly for all (x, y) on the line $y = x$, $f(x, y) = f(x, x) = \frac{1}{2}$,

$$\Rightarrow |f(x, 0) - f(x, x)| = \frac{1}{2} \text{ for all } x \in \mathbf{R}. \quad (1)$$

Take any $\epsilon \leq \frac{1}{4}$, say $\epsilon = \frac{1}{8}$.

Since for all $\delta > 0$, however small, there exists points (x, y) of the straight line $y = mx$

$$\text{such that } 0 < \sqrt{x^2 + y^2} = \sqrt{x^2 + m^2x^2} < \delta, \quad (2)$$

(1) and (2) implies that given $\epsilon = \frac{1}{8}$, there exists no $c \in \mathbf{R}$ and no $\delta > 0$,

$$\text{such that } |f(x, y) - c| < \epsilon = \frac{1}{8} \quad \text{if } 0 < \sqrt{x^2 + y^2} < \delta.$$

$$(b) \text{ Clearly } |x| \leq \sqrt{x^2 + y^2} \text{ and } |y| \leq \sqrt{x^2 + y^2}$$

$$\Rightarrow |x||y| \leq (x^2 + y^2)$$

$$\Rightarrow |f(x, y)| = \frac{|x||y|}{\sqrt{x^2 + y^2}} \leq \sqrt{x^2 + y^2}. \quad (1)$$

Given $\epsilon > 0$, take $\delta = \epsilon$, then from (1) it follows

$$|f(x, y) - 0| < \epsilon \quad \text{if} \quad 0 < \sqrt{x^2 + y^2} < \delta.$$

$$\Rightarrow \lim_{(x, y) \rightarrow (0, 0)} \frac{xy}{\sqrt{x^2 + y^2}} = 0.$$

(c) Clearly $|x^3| = |x| x^2 \leq \sqrt{x^2 + y^2}(x^2 + y^2)$ and $|y^3| = |y| y^2 \leq \sqrt{x^2 + y^2}(x^2 + y^2)$,

$$\begin{aligned} |f(x, y)| &= \left| \frac{x^3 - y^3}{x^2 + y^2} \right| \leq \left| \frac{x^3}{x^2 + y^2} \right| + \left| \frac{y^3}{x^2 + y^2} \right| \\ &\leq 2 \frac{\sqrt{x^2 + y^2}(x^2 + y^2)}{x^2 + y^2} = 2\sqrt{x^2 + y^2}. \end{aligned} \quad (1)$$

Given $\epsilon > 0$ take $\delta = \frac{\epsilon}{2}$ then from (1) it follows

$$|f(x, y) - 0| < \epsilon \quad \text{if} \quad 0 < \sqrt{x^2 + y^2} < \delta.$$

$$\text{Hence} \quad \lim_{(x, y) \rightarrow (0, 0)} \frac{x^3 - y^3}{x^2 + y^2} = 0.$$

(d) $f(x, y) = \frac{m}{1 + m^2}$, for all (x, y) such that $x = my^3$, $y \neq 0$.

Hence by a similar argument as in part(a),

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{xy^3}{x^2 + y^6} \text{ does not exist.}$$

(e) Note that $\lim_{(x, y) \rightarrow (0, 0)} x = 0$, and $\lim_{(x, y) \rightarrow (0, 0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = \lim_{r^2 \rightarrow 0} \frac{\sin r^2}{r^2} = 1$,

where $x = r \cos(\theta)$ and $y = r \sin(\theta)$.

$$\text{Hence} \quad \lim_{(x, y) \rightarrow (0, 0)} \frac{x \sin(x^2 + y^2)}{x^2 + y^2} = \left(\lim_{(x, y) \rightarrow (0, 0)} x \right) \times \left(\lim_{(x, y) \rightarrow (0, 0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} \right) = 0 \times 1 = 0.$$

Aliter: Since $|\sin(x^2 + y^2)| \leq (x^2 + y^2)$,

$$|f(x, y)| = \left| \frac{x \sin(x^2 + y^2)}{x^2 + y^2} \right| \leq |x| \leq \sqrt{x^2 + y^2}.$$

Given $\epsilon > 0$, take $\delta = \epsilon$, then

$$|f(x, y) - 0| < \epsilon \quad \text{if} \quad 0 < \sqrt{x^2 + y^2} < \delta.$$

(f) $\lim_{(x, y) \rightarrow (4, \pi)} x^2 = \lim_{x \rightarrow 4} x^2 = 16$ and $\lim_{(x, y) \rightarrow (4, \pi)} \sin\left(\frac{y}{x}\right) = \lim_{u \rightarrow \frac{\pi}{4}} \sin(u) = \sin\left(\frac{\pi}{4}\right)$

$$\Rightarrow \lim_{(x, y) \rightarrow (4, \pi)} x^2 \sin\left(\frac{y}{x}\right) = \left(\lim_{(x, y) \rightarrow (4, \pi)} x^2 \right) \times \left(\lim_{(x, y) \rightarrow (4, \pi)} \sin\left(\frac{y}{x}\right) \right) = 16 \sin\left(\frac{\pi}{4}\right) = 8\sqrt{2}.$$

15. Examine whether

$$f(x, y) = \begin{cases} \frac{xy(y^2 - x^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

is a continuous function.

Solution: Since $f(x, y)$ is of the form $f(x, y) = \frac{g(x, y)}{h(x, y)}$

where g, h are continuous functions and $h(x, y) \neq 0$ for all $(x, y) \neq (0, 0)$,

$f(x, y)$ is continuous at all $(x, y) \neq (0, 0)$.

To check the continuity of f at $(x, y) = (0, 0)$.

$$|f(x, y)| = \frac{|xy|(y^2 - x^2)|}{x^2 + y^2} \leq \frac{|xy|(y^2 + x^2)}{x^2 + y^2} \leq |xy| \leq (x^2 + y^2) \text{ for } (x, y) \neq (0, 0).$$

Given $\epsilon > 0$ take $\delta = \sqrt{\epsilon}$, then

$$|f(x, y) - 0| \leq (x^2 + y^2) < \epsilon \quad \text{if} \quad 0 < \sqrt{x^2 + y^2} < \delta,$$

hence $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.

Since $f(0, 0) = 0$, f is continuous at $(0, 0)$.

Extra Questions

1. Show that the scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ represents the volume of the parallelepiped determined by the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.
2. If $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$ are noncoplanar vectors, let

$$\mathbf{k}_1 = \frac{\mathbf{v}_2 \times \mathbf{v}_3}{\mathbf{v}_1 \cdot (\mathbf{v}_2 \times \mathbf{v}_3)}, \quad \mathbf{k}_2 = \frac{\mathbf{v}_3 \times \mathbf{v}_1}{\mathbf{v}_1 \cdot (\mathbf{v}_2 \times \mathbf{v}_3)} \quad \text{and} \quad \mathbf{k}_3 = \frac{\mathbf{v}_1 \times \mathbf{v}_2}{\mathbf{v}_1 \cdot (\mathbf{v}_2 \times \mathbf{v}_3)}.$$

Show that

(a) $\mathbf{k}_i$ is perpendicular to $\mathbf{v}_j$ if $i \neq j$

(b) $\mathbf{k}_i \cdot \mathbf{v}_i = 1$ for $i = 1, 2, 3$

(c) $\mathbf{k}_1 \cdot (\mathbf{k}_2 \times \mathbf{k}_3) = \frac{1}{\mathbf{v}_1 \cdot (\mathbf{v}_2 \times \mathbf{v}_3)}.$

3. Given the vectors $\mathbf{a} = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ and $\mathbf{b} = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$, verify that

$$|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2,$$

by computing each side in terms of the components of $\mathbf{a}$ and $\mathbf{b}$.

4. Show that the curvature of a plane parametric curve $x = f(t)$, $y = g(t)$ is

$$\kappa = \frac{|\dot{x}\ddot{y} - \dot{y}\ddot{x}|}{[\dot{x}^2 + \dot{y}^2]^{\frac{3}{2}}}$$

where the dots indicate the derivatives with respect to t .

Solution: $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$,

$\Rightarrow \mathbf{r}'(t) = \dot{x}(t)\mathbf{i} + \dot{y}(t)\mathbf{j}$, $\mathbf{r}''(t) = \ddot{x}(t)\mathbf{i} + \ddot{y}(t)\mathbf{j}$.

$\mathbf{r}'(t) \times \mathbf{r}''(t) = (x(t)\dot{y}(t) - y(t)\dot{x}(t))\mathbf{k}$.

$$\kappa(t) = \left| \frac{\mathbf{r}'(t) \times \mathbf{r}''(t)}{|\mathbf{r}'(t)|^3} \right| = \frac{|x(t)\dot{y}(t) - y(t)\dot{x}(t)|}{((\dot{x}(t))^2 + (\dot{y}(t))^2)^{\frac{3}{2}}}.$$

5. If $\mathbf{u}(t) = \mathbf{i} - 2t^2\mathbf{j} + 3t^3\mathbf{k}$ and $\mathbf{v}(t) = t\mathbf{i} + \cos t\mathbf{j} + \sin t\mathbf{k}$, find

(a) $D_t[\mathbf{u}(t) \cdot \mathbf{v}(t)]$

(b) $D_t[\mathbf{u}(t) \times \mathbf{v}(t)]$.

(c) $\lim_{t \rightarrow \pi} [\mathbf{u}(t) \cdot \mathbf{v}(t)] = \lim_{t \rightarrow \pi} [\mathbf{u}(t)] \cdot \lim_{t \rightarrow \pi} [\mathbf{v}(t)]$ and $\lim_{t \rightarrow \pi} [\mathbf{u}(t) \times \mathbf{v}(t)] = \lim_{t \rightarrow \pi} [\mathbf{u}(t)] \times \lim_{t \rightarrow \pi} [\mathbf{v}(t)]$